

Setup + first drive

THE BIG IDEA

Get every tool installed and run your first robot code.

Everything else in the curriculum builds on what you set up today.

KEY CONCEPTS — LESSON 01

WPILib + VS Code Editor	AdvantageScope Dashboard	First Drive project Code base	Simulator Test bench	Reference card Cheat sheet
The FRC editor. Java + libraries. Purple = correct.	Bundled with WPILib. Real-time graphs. 2D/3D field view.	Complete working robot. Build with Ctrl+Shift+P. BUILD SUCCESSFUL = good.	Runs code on laptop. No hardware today. Drive with WASD.	Two pages of patterns. Methods + conversions. Look up, then ask.

THE SETUP WORKFLOW

- 1 Run WPILib installer → launch WPILib VS Code
- 2 Extract Lesson01-Project.zip → open Lesson01\First Drive → Build
- 3 Ctrl+Shift+P → Simulate Robot Code → check halsim_gui
- 4 Drag Keyboard 0 to Joystick[0] → enable Teleoperated
- 5 Ctrl+Shift+P → WPILib: Start Tool → AdvantageScope
- 6 File → Connect to Simulator → NetworkTables 4 → time bar starts
- 7 Line Graph → Drive → drag Left/RightVelocityMetersPerSec → drive
- 8 2D Field → RealOutputs → Drive → drag Robot (Pose2d) → drive
- 9 3D Field → add Robot + FinalComponentPoses → Custom Assets → restart → XRP Robot → W/S/A/D, arm Z/X

AFTER THIS LESSON I CAN...

- ☐ Open WPILib VS Code (purple theme)
- ☐ Launch AdvantageScope via WPILib: Start Tool
- ☐ Build the First Drive project to BUILD SUCCESSFUL
- ☐ Drive a simulated robot with W, S, A, D
- ☐ Connect AdvantageScope and graph Left/RightVelocityMetersPerSec
- ☐ View the robot moving in AdvantageScope's 2D field view
- ☐ View robot and arm in 3D field view with the XRP robot model
- ☐ Find the reference card and locate its main sections

QUESTIONS I STILL HAVE

Write your questions here. Bring them to the next session.

KEY VOCABULARY

WPILib — The official FRC software suite — VS Code + Java + libraries + tools + AdvantageScope in one installer

AdvantageScope — Log viewer for robot data — real-time graphs, 2D/3D field views, replay from saved logs

Simulator — Runs robot code on your laptop with no hardware — same Java logic as a real robot

Build — Compiles Java source files into runnable bytecode — must succeed before you simulate

Reference card — Two-page printable cheat sheet — XRP methods, unit conversions, the patterns you'll forget

Write anything here — surprises, connections, things to look up later.